

Web Technology Presentation

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Section: 15



Scenario 2

Incident Reporting System



Introduction

Background

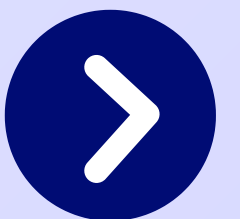
DHL as a logistic-focused company receive many incident reports regarding involving late deliveries, damaged parcels, address issues, system errors, and customer complaints. from various different channels every day.

Problem Statement

DHL Management teams face difficulties generating reports and monitoring incident status because the information is not centralized.

Proposed Solution

A web-based system for uploading, managing, searching, and tracking incident reports as a centralized platform.



Features

Incident Reporting System has the following features.

1 Authentication & Authorization

User must login first before able to access the system

2 File Upload

User can upload incident files in form of .txt files

3 Manage Incident

User can update or delete existing incident record

4 Search & Filter

Search, filter and sort through the incident list

5 Dashboard Statistics

Dashboard shows the statistics of all incident statuses



Architecture

Architecture explains the technology, programming language and software to develop the system.

Front-End

Front-end defines the structure and styling of the website

HTML, CSS, JavaScript

Back-End

Back-end defines how the data is processed and stored

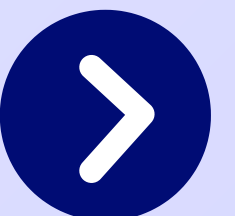
JavaScript as APIs, JSON

Software/Library

VS Code: To code the program files

Node.JS: To locally host the server in port 3000

Google Chrome (Live Server): To dynamically view the system



HTML Files

Each HTML file represents a webpage in the system.

1) login.html

Login page where user authenticates the login credentials.

2) main.html

Dashboard page where statistics are displayed and user can navigate to listing or to upload page.

3) upload.html

Upload page where user uploads new incident report.

4) logs.html

Listing page where all incident records are displayed in ordered manner.

5) info.html

Detailed information of the incident is displayed and user can edit or delete the incident.



Application Programming Interfaces (APIs)

APIs here define the logic for each HTML page.

1) loginAPI.js

API to process login credentials

2) mainAPI.js

API to process dashboard statistics

3) uploadAPI.js

API to process file upload logic

4) logsAPI.js

API to process all the incident listing logics including searching, sorting and filtering

5) infoAPI.js

API to process the detailed incident record including data edit and deletion



Database Structure

All the data of the system is stored in db.json file. In the file, only 2 json objects are present;

users

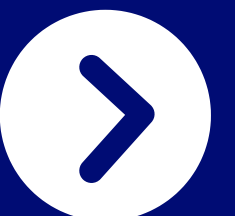
Stores all the user credentials for authentication & authorization

- id
- username
- password

incidents

Stores all the incident details

- id
- creator
- date
- status
- description
- tag
- department
- channel



Demonstration

Note: No RPA is implemented due to technical issues.

